Skin Cancer Detection Using Advanced DeepLearning

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Abstract—There are more than 200 types of cancer and we can analyze them according to where they start in our body. So,Melanoma is examined as one of the skin cancer which causes death in a severe manner all over the world.If it is not diagnosed in early stage then there is a chance of spreading to other body parts. In this paper, we performed skin cancer using deep learning models by considering a benign lesion. Medical imaging is today one of the main tools for diagnosing cancers. It allows us to obtain precise images, internal organs and thus to visualize the possible tumours that they present.We had implemented our project by using deep learning models to get precise classification between melanoma and non melanoma skin disease.

Index Terms—

1 INTRODUCTION

Cancer is a serious and life-threatening disease .It can occur in any part of the body, there are many different types of cancer may exist in human body and skin cancer is one of the fastest-growing cancers that can cause death [**?**].Skin cancer is defined as the uncontrolled growth of cells in the skin. It most often develops on areas of the skin exposed to the sun's rays. In the US, more people are diagnosed with skin cancer each year . However, skin cancer is also the easiest cancer to be treated , especially when detected early. If Skin cancer is not detected in early stages, it cannot be cured.

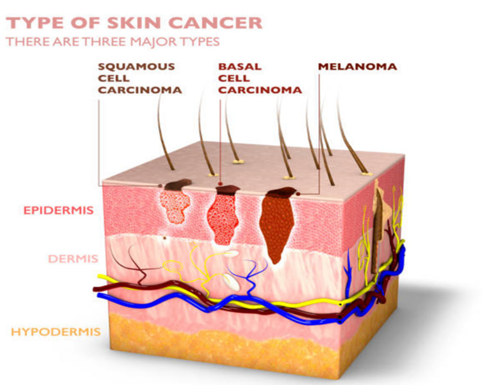

Fig. 1. Types of Skin cancer

Skin cancer can be mainly categorized as three types such as Basal cell carcinoma(BCC), Melonoma and Squamous cell carcinoma (SCC). Skin Cancer can be divided into two categories, Melonoma and Non-Melonoma.The non-melonomas were BCC and SCC. Basal cell cancer grows slowly and can damage the tissue around it but are unlikely to spread to distant areas . Squamous cell cancer is more likely to spread .The melanoma is the most serious form of the skin cancer because it is more likely to spread to other parts of the body. Once melanoma spreads beyond the skin to other parts of the body, it becomes hard to treat. [] Melonoma skin responsible for pigmentation known as melanocytes.Melonoma is caused due to presence of Melonocytes which are present with in the body. According to the World Health Organization (WHO), skin cancer is one of the most commonly diagnosed cancers, with more than 1.5 million cases diagnosed annually and over 120 000 skin cancer-associated deaths were reported. Skin cancers are caused primarily by exposure to ultraviolet radiation (UVR), either from the sun or from artificial sources such as sunbeds.To detect skin cancer, a visual inspection of the skin is often performed by a doctor or dermatologist. In some cases, a biopsy of the affected skin may be taken to confirm the diagnosis. Additionally, there are several technological tools and methods that can be used to detect skin cancer, including dermoscopy, computerized analysis of images, and the use of ultraviolet (UV) and visible light imaging. . The success of skin cancer depends on early diagnosis and appropriate treatment . In diagnosing, dermatologists typically will follow the mnemonic "ABCDE" when looking at specific moles. The five important variables in looking for melanoma include: asymmetry, borders, color, diameter, and evolution over time.

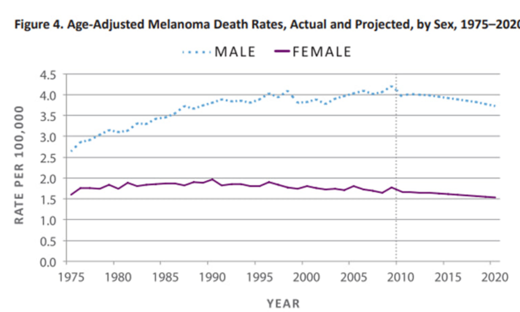

Fig. 2. Melonoma Death rate

Melanoma incidence and death rates are highest among males. Men aged 65 years or older have the highest incidence (130.1 cases per 100,000) and death rates (23.7 per 100,000) for melanoma.

1.1 Deep Learning:

DEEP LEARNING is a rapidly developing area of research that involves the use of artificial neural networks to detect skin cancer.The algorithms have the ability to learn from large amounts of data and can accurately identify patterns and features in medical images that are often difficult for human experts to detect. Studies have shown that deep learning algorithms can achieve high accuracy rates in detecting skin cancer. It's important to note that early detection and prompt treatment can greatly improve the chances of successful treatment and recovery.

1.2 Convolutional Neural Network:

CNNs are inspired by the structure and function of the human brain, specifically the visual cortex. They consist of multiple layers of interconnected nodes that apply filters to the input image, which helps in feature extraction and classification. The CNN is used to categorise images and create a digit identification system. The architecture of a CNN typically consists of three types of layers: convolutional layers, pooling layers, and fully connected layers. Convolutional layers apply filters to the input image, which helps in feature extraction. Pooling layers reduce the spatial size of the output from the convolutional layer, which helps in reducing computation time and preventing overfitting. Fully connected layers perform the final classification based on the extracted features

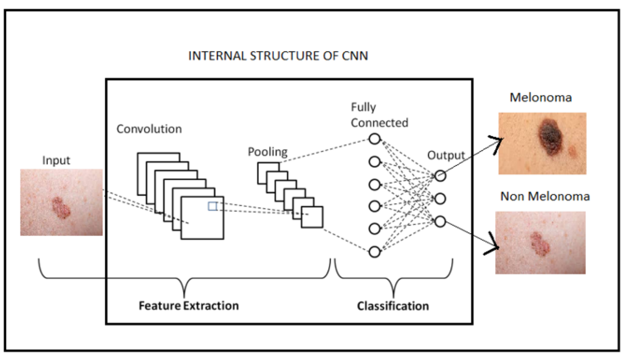

Fig. 3. Skin cancer using CNN method

2 Dataset used:

We used the publicly available Kaggle dataset of skin lesions obtained from the ISIC archive for model training and validation (International Skin Image Collection). There are 1800 benign samples (1440+360) and 1497 malignant samples (1197+300).Validation data is set to 30percent of test data. 220 images for validation and 440 images for testing And the training set, because there are 2637 images. That signifies that nearly 5percent of total data is used for validation, 10percent for testing, and 85percent for training.

3 Literature Review:

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